

Supplementary Information

Reduced molecular dynamics trajectories of 207 active GPCR-G-protein complexes

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Supplementary Note 1. Frozen cohort and the 208-to-207 transition

The authoritative cohort is [data/release_cohort_v9_final207.csv](#). It contains 207 unique [system_id](#) values and is used as an allow-list for all v12 tables, figures and API consistency checks. The class counts are 181 Class A and 26 Class B. The family counts are 95 Gi/o, 65 Gs, 41 Gq/11 and 6 G12/13. The cross-tabulation is:

GPCR class	Gi/o	Gs	Gq/11	G12/13	Total
Class A	92	46	39	4	181
Class B	3	19	2	2	26
Total	95	65	41	6	207

The 222-record working inventory is partitioned into 207 included, 13 unresolved and two excluded records. The two exclusions are:

- [Gq_7E9W](#): non-GPCR duplicate or mislabel of [Gq_8E9W](#).
- [Gq_7DWC](#): duplicate source identity of [Gq_8DWC](#); both source records were assigned to MRGPRX1 (Q96LB2).

Earlier 208-system files included [Gq_7DWC](#). Its exclusion changes 208 systems to 207, Class A from 182 to 181, Gq/11 from 42 to 41, production repeats from 624 to 621 and nominal sampling from 312.0 to 310.5 μ s. Historical files containing [final208](#) or [final624](#) in their names are source-provenance inputs only. Their contents are filtered against the final-207 allow-list, and no active v12 submission artifact uses those labels.

[Gs_8HTI](#) is retained as a valid system but has no canonical receptor UniProt accession. It is serialized with an explicit null value, not a fabricated accession.

Supplementary Note 2. Source-protocol evidence

The protocol assignment is regenerated by joining the final cohort to [parallel_codex/provenance_direct/protocol_system_evidence.csv](#). The counts are P1 179, P2 26 and P3 2. There are 205 AMBER-family systems and two GROMACS systems. Source records include production output paths, hashes where available, recovered version strings, output interval, random-seed evidence and a structured settings record.

Evidence is classified as [verified](#) for 202 systems and [partial](#) for five systems:

System	Protocol	Evidence status	Interpretation
Gi_8EF0	P1	partial	one or more requested provenance fields were not recovered

The v12 text therefore does not assign one global engine version to the collection. Recovered version strings and their evidence scope are retained per system in Supplementary Data S7. Protocol group, construct provenance and source PDB should be treated as potential batch variables.

The three production replicas are computational repeats. They do not represent biological specimens, independent receptor constructs or independent experimental structures.

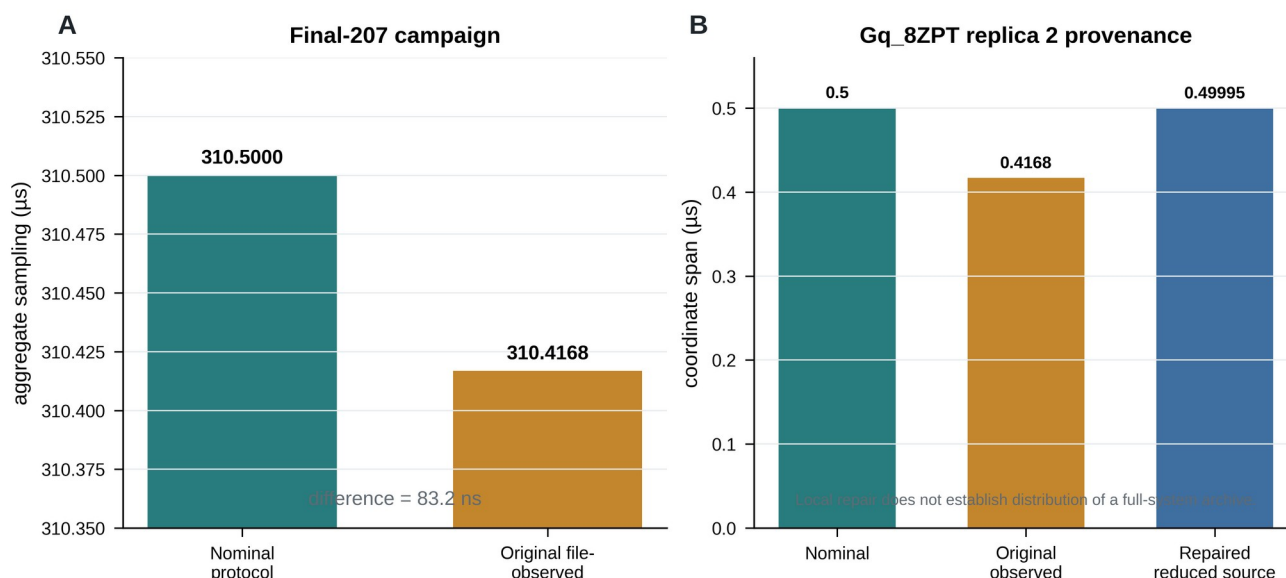
Supplementary Note 3. Sampling definitions and Gq_8ZPT replica 2

Three sampling quantities must remain distinct:

1. **Nominal protocol sampling:** $207 \times 3 \times 500 \text{ ns} = 310.5 \text{ } \mu\text{s}$.
2. **Original file-observed coordinate sampling:** $310.4168 \text{ } \mu\text{s}$.
3. **Reduced-release coordinate coverage:** the represented time between the first and last of 2,500 frames is 499.8 ns per released record.

The 83.2-ns difference between the first two quantities arises from the original Gq_8ZPT replica 2 coordinate record, which contains 416.8 ns rather than 500 ns. A continuation-derived coordinate source used for reduced-release repair was independently checked over 10,000 frames. It has monotonic timestamps, no non-finite frame, no invalid box frame and a represented span of 499.95 ns. The audit verdict is **GOOD**.

The repaired source is sufficient for deterministic production of the reduced PDB/XTC record. It is not evidence that a repaired full-system trajectory, topology and preparation input set are deposited. The v12 manuscript reports nominal and original file-observed campaign totals separately.



Supplementary Figure S2 | Nominal, original file-observed and repaired reduced-source sampling. (A) The final-207 nominal protocol design is $310.5 \text{ } \mu\text{s}$; original coordinate files contain $310.4168 \text{ } \mu\text{s}$. (B) Gq_8ZPT replica 2 accounts for the difference. The continuation-derived reduced source spans 499.95 ns. This local repair does not establish public distribution of a repaired full-system archive.

Supplementary Note 4. Reduced-record construction

The replica-specific release builder receives a replica number and the frozen 207-system table. Each target system directory contains a copied canonical `structure.pdb` and a generated `traj.xtc`. The PDB defines the retained atoms and ordering.

For a source trajectory already matching the PDB, coordinates are written directly. When a full-system or differently ordered source contains additional atoms, target atoms are mapped by (`resname`, `resid`, `atom name`) against a matching source topology. This selection is checked to contain exactly the PDB atom count. The reduction does not infer a ligand-selection rule; users must inspect the released PDB to determine which non-protein components, if any, were retained.

The output schedule is:

Source form	Selection	Output
10,000 or 10,001 frames at ~50 ps	every fourth frame, first 10,000-frame span	2,500 frames
2,501 frames at ~200 ps	first 2,500 frames	2,500 frames
canonical 2,500-frame reduced source	preserve	2,500 frames

All released trajectories use an approximately 200-ps interval and represent 499.8 ns between first and last frame. The reduction excludes membrane lipids, solvent and mobile ions. The underlying full-system trajectories are not included in the reduced release.

Supplementary Note 5. Full-frame reduced-release QC

The JSON audit for each release replica contains one `system_qc` record per system and an aggregate status. Every frame of each locally complete release is tested for finite coordinates, monotonic finite time and a valid unit cell. The PDB/XTC atom count must match. Reference backbone geometry is used to detect atom-order errors while documented starting-coordinate gaps are excluded from false-positive bond checks. Component geometry is screened for catastrophic chain breaks, coordinate scatter and receptor–G-protein separation.

The release-level assertions are:

Assertion	Required result
Cohort directories	exactly 207
Unresolved or excluded directories	0
PDB files	207
XTC files	207
Frames per XTC	2,500
Source replica values	one declared replica
Manifest absolute paths	0
PDB/XTC atom-count mismatches	0
Non-finite coordinate frames	0
Non-monotonic time records	0
Invalid periodic boxes	0
Catastrophic chain-break systems	0
Coordinate-scatter systems	0
Complex-separation systems	0

At the v12 build time, replicas 1 and 3 satisfy the local manifest and full-frame-QC requirements. Replica 2 remains under parallel generation and is represented as pending, not as a failed or zero-sized record. The three remote records remain private drafts. The v12 release gate must be rerun after replica 2 completes.

Supplementary Note 6. Independent portal-oriented reduced-record audit

The portal-oriented reduced records predate the uniform replica-specific release builder and have a nonuniform frame-count distribution. The exact final-207 audit samples 40 frames per system and checks:

- PDB/XTC atom-count agreement;
- PDB element-field completeness;
- presence and consistency of unit-cell dimensions;
- intra-chain backbone breaks;
- coordinate scatter; and

- inter-chain splitting.

All 207 systems are classified **OK**. The frame-count distribution is $203 \times 2,500$, $2 \times 2,501$, 1×681 (**Gq_7RAN**) and 1×200 (**Gs_3SN6**). These counts describe the older portal-oriented records. They do not describe the three replica-specific reduced releases, for which the target is uniformly 2,500 frames.

Supplementary Note 7. Pocket and gateway records

Pocket records

Supplementary Data S5 has 2,149 detected-pocket rows from 205 systems. Two additional rows explicitly record valid completed analyses with no persistent pocket: **Gi_7YK6** and **Gi_8YIC**. The system denominator is therefore:

```
205 systems with detected pockets
+ 2 systems with valid zero-pocket results
= 207 system records
```

The 58-system orthosteric positive-control denominator is defined by the presence of a peptide-based reference and an evaluable pocket analysis. Forty-nine systems recover an orthosteric cluster at the specified 0.85 frequency threshold. Ligand-free systems are not counted as failures. The analysis is a technical check on annotation recovery; pocket recurrence, ranking, biological interpretation and druggability are outside the database paper.

Gateway records and reproducibility boundary

Supplementary Data S6 contains 5,796 rows:

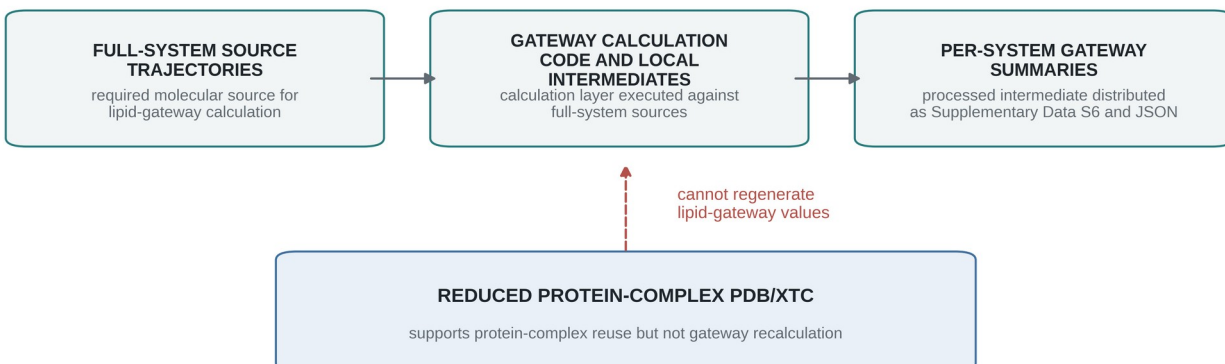
```
207 systems × 7 adjacent TM interfaces × 4 metrics
```

The values were calculated from full-system trajectories containing membrane lipids. The reduced release contains no membrane lipids. S6 therefore serves as the processed intermediate needed to reproduce database-paper summaries, but the molecular calculation cannot be rerun from the reduced XTC files. The active API consensus cache has been regenerated from the final-207 intermediate:

Group	n_systems
ALL	207
Class A	181
Class B	26
Gi/o	95
Gs	65
Gq/11	41
G12/13	6

The v12 main figures do not display the family-organized gateway distribution.

Gateway-summary provenance and the reduced-record reproducibility boundary



Supplementary Figure S1 | Gateway-summary provenance and reduced-record boundary. Full-system trajectories are required to calculate lipid-gateway records. The calculation code produces per-system processed summaries distributed in Supplementary Data S6 and the API. Reduced PDB/XTC records support protein-complex reuse but cannot regenerate lipid-gateway measurements.

Supplementary Note 8. Worked checksum and loading example

The following complete example reads the replica-1 manifest, locates [G12_7SF7](#), verifies both SHA-256 digests and loads the PDB/XTC pair. The user must first extract the relevant family archive without changing member names.

```

from pathlib import Path
import hashlib
import pandas as pd
import MDAnalysis as mda

release = Path(".")
manifest = pd.read_csv(
    release / "CoupledMD_reduced_trajectory_manifest.csv"
)
row = manifest.loc[
    manifest["system_id"].eq("G12_7SF7")
    & manifest["source_replica"].eq(1)
].iloc[0]

pdb_path = release / row["pdb_relative_path"]
xtc_path = release / row["xtc_relative_path"]

def sha256(path):
    digest = hashlib.sha256()
    with path.open("rb") as handle:
        for block in iter(lambda: handle.read(8 * 1024 * 1024), b''):
            digest.update(block)
    return digest.hexdigest()

assert pdb_path.is_file()
assert xtc_path.is_file()
assert sha256(pdb_path) == row["pdb_sha256"]
assert sha256(xtc_path) == row["xtc_sha256"]

universe = mda.Universe(pdb_path, xtc_path)
assert universe.atoms.n_atoms == int(row["n_atoms"])
assert len(universe.trajectory) == int(row["n_frames"]) == 2500
assert float(row["frame_interval_ps"]) == 200.0
assert float(row["represented_span_ns"]) == 499.8

print(
    row["system_id"],
    universe.atoms.n_atoms,
    len(universe.trajectory),
)

```

The same verification can be performed without Python:

```

sha256sum path/from/manifest/structure.pdb
sha256sum path/from/manifest/traj.xtc

```

Both output strings must match the manifest exactly before analysis.

Supplementary Note 9. Reproducible v12 commands

The v12 build is regenerated from the manuscript workspace with:

```

python audit_v12_release_status.py
python update_v12_gateway_consensus.py
python run_final207_viz_audit.py
python build_v12_data.py
MPLCONFIGDIR=/tmp/mpl-v12 python plot_v12_figures.py
python build_v12_documents.py
python validate_v12_submission.py

```

`run_final207_viz_audit.py` is the active wrapper around a historical implementation that predates the 207-system freeze. Its output is `final207_reduced_visualization_audit_40frames.csv`. The historical implementation filename is not distributed as an active v12 artifact. All generated source-data tables and reports use v12 or final-207 naming.

The frozen environment and file hashes are in Supplementary Data S10 and [reports/v12_environment_freeze.json](#). The build must be rerun after any release-status change because remote-state fields, release manifests, QC tables, Figure 5 and Data Availability are coupled.

Supplementary Table S1. Molecular record specification

Record	Format	Pairing	Molecular scope	Supported use	Explicit limitation
Reduced structure	PDB	reduced XTC	atoms defined by PDB	topology and atom-order reference	not a full-system topology
Reduced trajectory	XTC	reduced PDB	selected protein-complex representation	protein-complex structural reuse	no membrane, solvent, mobile ions or fast kinetics
System metadata	CSV/JSON	system_id	207 systems	discovery and joins	no coordinates
Replica provenance	CSV	system_id , replica_id	621 expected repeats	protocol and sampling audit	source molecular files not included
Pocket summary	CSV/JSON	system_id	207 completed system records	technical annotation reuse	no experimental druggability claim
Gateway summary	CSV/JSON	system_id , interface, metric	processed full-system result	reproduction of reported summaries	cannot be recalculated from reduced XTC
Release QC	CSV/JSON	system and release replica	reduced files	integrity screening	does not validate unavailable full-system files

Supplementary Table S2. Database-versus-resource claim and data overlap

Topic or artifact	Database Data Descriptor	Resource-atlas manuscript	Permitted overlap and disclosure
207-system cohort and 222-record boundary	owns definition, exclusions and identifiers	may cite as analysis cohort	exact metadata may be reused with citation
Simulation construction and protocols	owns provenance and limitations	may summarize only as input provenance	methods overlap must be disclosed
Three computational repeats	owns record definition, sampling truth and QC	may use as analysis inputs	must not call biological replicates
Reduced PDB/XTC release	owns archive, checksums and reuse	may link to data	no full-system claim
Pocket calculation method	owns limited method and positive control	may reuse method	shared processed inputs disclosed
2,149 pockets; 205 detected + 2 zero	owns completeness accounting	may use as atlas input	database paper does not rank or interpret
Orthosteric recovery 49/58	owns technical positive control	may cite validation	no druggability conclusion
Recurrent pocket clusters and rankings	excluded	owns	no main-text or figure overlap
Sixty-five pocket clusters	excluded	owns	absent from database figures and headlines
C7–Gq enrichment	excluded	owns	absent
Partner-associated pocket changes	excluded	owns	absent
Gateway biological distributions	method/intermediate only; no main figure	owns biological comparison	S6 shared as disclosed input
Gateway biological interpretation	excluded	owns	absent
$\alpha 5$ or family-selectivity analyses	excluded	owns	absent
Mechanistic or druggability conclusions	excluded	owns	absent
Learned models, predictions and BRET	excluded	excluded from resource paper if reserved for AI study	no overlap in this descriptor

Supplementary Table S3. Current release and submission gate

Requirement	Current v12 state	Status
Final cohort is exactly 207 systems	machine verified	PASS
Three expected repeats per system	621-row design ledger	PASS
Exact final-207 portal-reduced audit	207/207 OK	PASS
Replica 1 local manifest and full-frame QC	207/207, passed	PASS
Replica 2 local manifest and full-frame QC	processing in parallel	HOLD
Replica 3 local manifest and full-frame QC	207/207, passed	PASS

Requirement	Current v12 state	Status
Three authenticated remote records	private, restricted, unsubmitted drafts	HOLD
Three unauthenticated DOI resolutions	not publicly resolvable at audit time	HOLD
Anonymous reviewer access	not machine verified	HOLD
Reduced-release component scope	explicitly documented	PASS
Stable full-system molecular access	absent	BLOCKED for full-system-only claims
Figure source data	generated for all panels	PASS
Gateway processed intermediates	S6 generated	PASS
Gateway recalculation from reduced XTC	impossible by design	BLOCKED if claimed
Paper repository v12 commit/tag	not yet pushed	HOLD

Supplementary Data files

File	Rows expected at v12 build	Description
Supplementary_Data_S1_included_system_inventory.csv	207	Frozen included cohort
Supplementary_Data_S2_release_boundary_exceptions.csv	15	Thirteen unresolved and two excluded records
Supplementary_Data_S3_metadata_dictionary.csv	variable	Field-level schema
Supplementary_Data_S4_reduced_release_manifest.csv	1,242	207 × 3 × two molecular file roles; pending entries retained
Supplementary_Data_S5_pocket_summaries_gpcrdb.csv	2,151	Detected and explicit zero-pocket records
Supplementary_Data_S6_gateway_per_system.csv	5,796	Processed gateway intermediates
Supplementary_Data_S7_source_simulation_replica_provenance.csv	621	Nominal, observed and repair provenance
Supplementary_Data_S8_reduced_release_qc.csv	621	Expected release QC grid
Supplementary_Data_S9_api_access_audit.csv	variable	API coverage and consensus-cache audit
Supplementary_Data_S10_code_environment_inventory.csv	variable	Build files and software versions

Supplementary Data Availability

The three Zenodo identifiers are reserved for replica-specific reduced releases:

- Replica 1: [10.5281/zenodo.21395292](https://zenodo.org/record/21395292)
- Replica 2: [10.5281/zenodo.21447748](https://zenodo.org/record/21447748)
- Replica 3: [10.5281/zenodo.21448037](https://zenodo.org/record/21448037)

At the v12 audit time, all are restricted, unsubmitted private drafts and none resolves publicly without authentication. Replica 2 remains under processing. No DOI, public-release or archive-completeness claim should be copied from this supplement until the remote audit is rerun and all records pass.

The code/web repository is <https://github.com/HuangJianxiang-SJTU/coupledmd>. The paper and source-data repository is https://github.com/HuangJianxiang-SJTU/coupledmd_database. A v12 commit and release tag remain to be created. The underlying full-system trajectories are not included in the reduced Zenodo records.